WORKED SOLUTION

STEP III 2024, Question 11

Probability and Statistics • Binomial Distribution Expectation

STEP III 2024, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

The identity in (i) pairs each term $r\binom{2n}{r}$ in the bottom half of the sum with a term in the top half. That symmetry is exactly what X needs: the larger of heads and tails is either the number of heads or the number of tails, and swapping the roles of heads and tails folds the expectation onto the top half of the sum. After that, (iii) is a ratio test, and (iv) needs only a line.

Solution

Part (i)

For $1 \leq r \leq 2n$, cancelling r against $r!$ gives

$$r\binom{2n}{r} = \frac{r\,(2n)!}{r!\,(2n-r)!} = \frac{(2n)!}{(r-1)!\,(2n-r)!}.$$

On the other side, $2n-(2n+1-r) = r-1$, so

$$(2n+1-r)\binom{2n}{2n+1-r} = \frac{(2n+1-r)\,(2n)!}{(2n+1-r)!\,(r-1)!} = \frac{(2n)!}{(2n-r)!\,(r-1)!},$$

again cancelling the top factor against the factorial. (We need $1 \leq r \leq 2n$ so that $2n+1-r$ lies between 1 and $2n$, which makes both factorials sensible.) The two expressions agree, so the identity holds.

For the sum, the $r = 0$ term is zero. Split the rest into $1 \leq r \leq n$ and $n+1 \leq r \leq 2n$. In the lower block, put $s = 2n+1-r$. As r runs from 1 to n, s runs from $2n$ down to $n+1$, and by the identity

$$\sum_{r=1}^{n} r\binom{2n}{r} = \sum_{r=1}^{n}(2n+1-r)\binom{2n}{2n+1-r} = \sum_{s=n+1}^{2n} s\binom{2n}{s}.$$

So the lower block equals the upper block, and

$$\sum_{r=0}^{2n} r\binom{2n}{r} = 2\sum_{r=n+1}^{2n} r\binom{2n}{r},$$

as required.

Part (ii)

Let H be the number of heads, so $H \sim \mathrm{B}(2n, \frac{1}{2})$ and $\mathrm{P}(H = r) = \binom{2n}{r}/2^{2n}$. The number of tails is $2n - H$, so $X = \max(H, 2n - H)$. That is, $X = H$ when $H \geq n+1$, $X = n$ when $H = n$, and $X = 2n - H$ when $H \leq n-1$. Hence

$$2^{2n}\,\mathrm{E}(X) = \sum_{r=n+1}^{2n} r\binom{2n}{r} + n\binom{2n}{n} + \sum_{r=0}^{n-1}(2n-r)\binom{2n}{r}.$$

In the last sum put $s = 2n - r$, which runs from $2n$ down to $n+1$. Since $\binom{2n}{r} = \binom{2n}{2n-r}$, it becomes $\sum_{s=n+1}^{2n} s\binom{2n}{s}$, the same as the first sum. (This is the heads/tails symmetry: having s tails is as likely as having s heads.)

> **Ask yourself** How do we evaluate $\sum_{r=n+1}^{2n} r\binom{2n}{r}$, which only runs over half the range?

This is what part (i) is for. Its full-range version is given: with $2n$ in place of n, $\sum_{r=0}^{2n} r\binom{2n}{r} = 2n \cdot 2^{2n-1}$. So by (i),

$$2\sum_{r=n+1}^{2n} r\binom{2n}{r} = 2n \cdot 2^{2n-1} = n\,2^{2n}.$$

Therefore

$$2^{2n}\,\mathrm{E}(X) = n\,2^{2n} + n\binom{2n}{n}, \qquad \text{so} \qquad \mathrm{E}(X) = n\left(1 + \frac{1}{2^{2n}}\binom{2n}{n}\right),$$

as required.

Part (iii)

Let $a_n = \binom{2n}{n}/2^{2n}$, which is positive. Then

$$\frac{a_{n+1}}{a_n} = \frac{(2n+2)!}{(n+1)!\,(n+1)!} \cdot \frac{n!\,n!}{(2n)!} \cdot \frac{1}{4} = \frac{(2n+2)(2n+1)}{4(n+1)^2} = \frac{2n+1}{2n+2},$$

using $2n + 2 = 2(n+1)$. This ratio is less than 1, and $a_n > 0$, so $a_{n+1} < a_n$ for every $n \geq 1$. The sequence is strictly decreasing.

Part (iv)

Your winnings are exactly the X of part (ii), and you pay £n, so the expected winnings per pound paid are

$$\frac{\mathrm{E}(X)}{n} = 1 + \frac{1}{2^{2n}}\binom{2n}{n} = 1 + a_n.$$

By (iii), a_n is largest when n is smallest, so you should choose $n = 1$. Then you get back $1 + \frac{1}{2} = \frac{3}{2}$ pounds per pound paid, an expected profit of 50p per pound. (If "winnings" means profit, $\mathrm{E}(X)/n - 1 = a_n$, and the conclusion is the same.)

Checks

- For $n = 1$ (two tosses), $X = 2$ for HH or TT and $X = 1$ for HT or TH, so $\mathrm{E}(X) = \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 1 = \frac{3}{2}$. The formula gives $1 \cdot (1 + \frac{2}{4}) = \frac{3}{2}$.
- For $n = 2$, the formula gives $2(1 + \frac{6}{16}) = \frac{11}{4}$, so the return per pound is $\frac{11}{8} = 1.375 < 1.5$. For $n = 3$ it is $\frac{21}{16} = 1.3125$, still decreasing as (iii) predicts.
- The formula for $\mathrm{E}(X)$ was confirmed for $n \leq 5$ by a computer check over all 2^{2n} sequences of tosses.

Common slips

- Counting the $H = n$ term twice, or not at all. It sits on neither side of the split and contributes $n\binom{2n}{n}$ once.
- In (iv), maximising $\mathrm{E}(X)$ itself, which grows without limit, rather than $\mathrm{E}(X)$ per pound paid.

Beyond STEP

The excess $\mathrm{E}(X) - n = na_n$ is the expected distance of a simple random walk from its start, halved. By Stirling's formula, $a_n \approx 1/\sqrt{\pi n}$, so this excess grows like $\sqrt{n/\pi}$: the typical imbalance between heads and tails grows like the square root of the number of tosses. Meanwhile the excess per pound, a_n, tends to 0, which is the law of large numbers at work. Grimmett and Welsh, *Probability: An Introduction*, covers random walks and these limits.

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